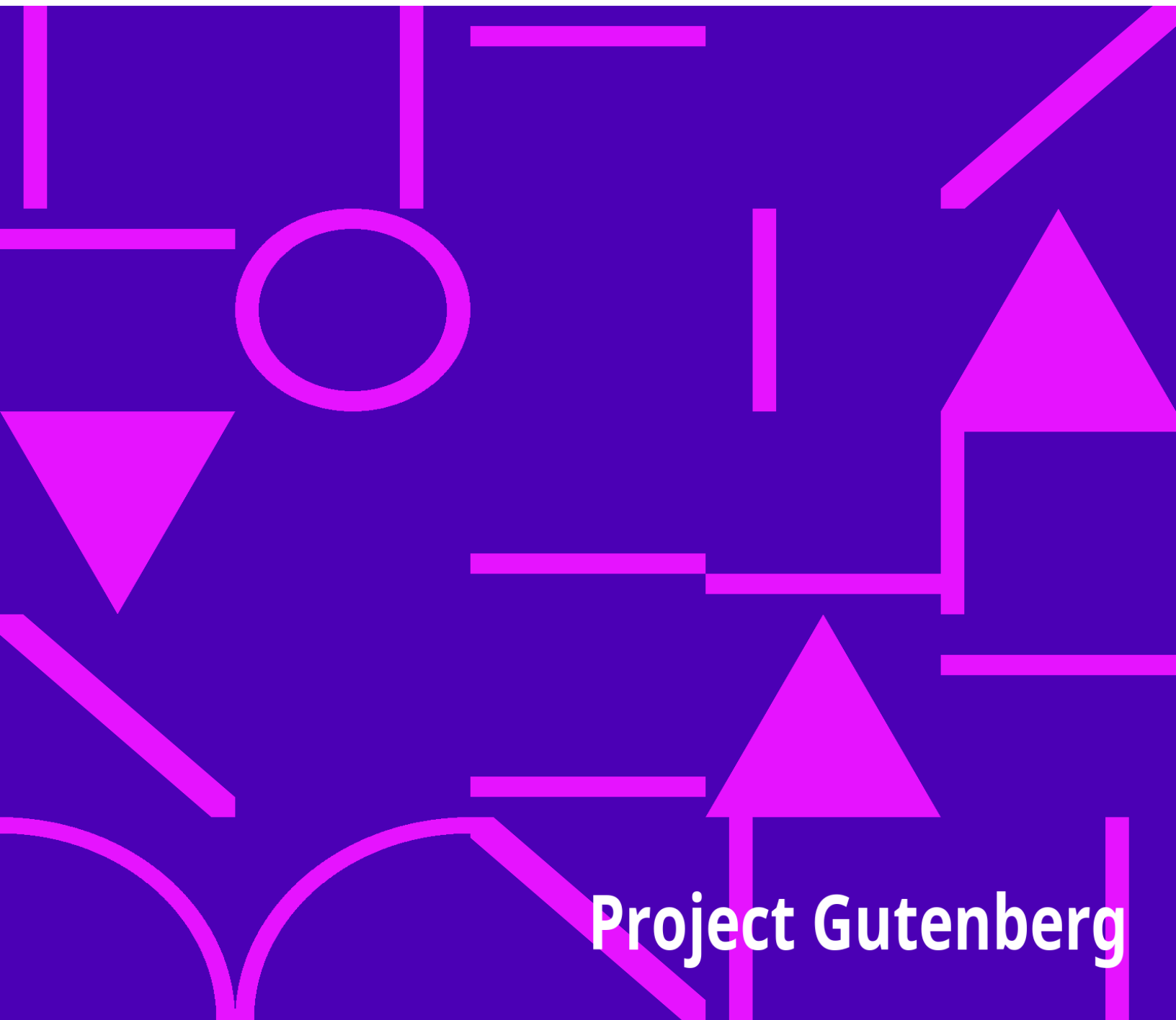


The new gas from radium

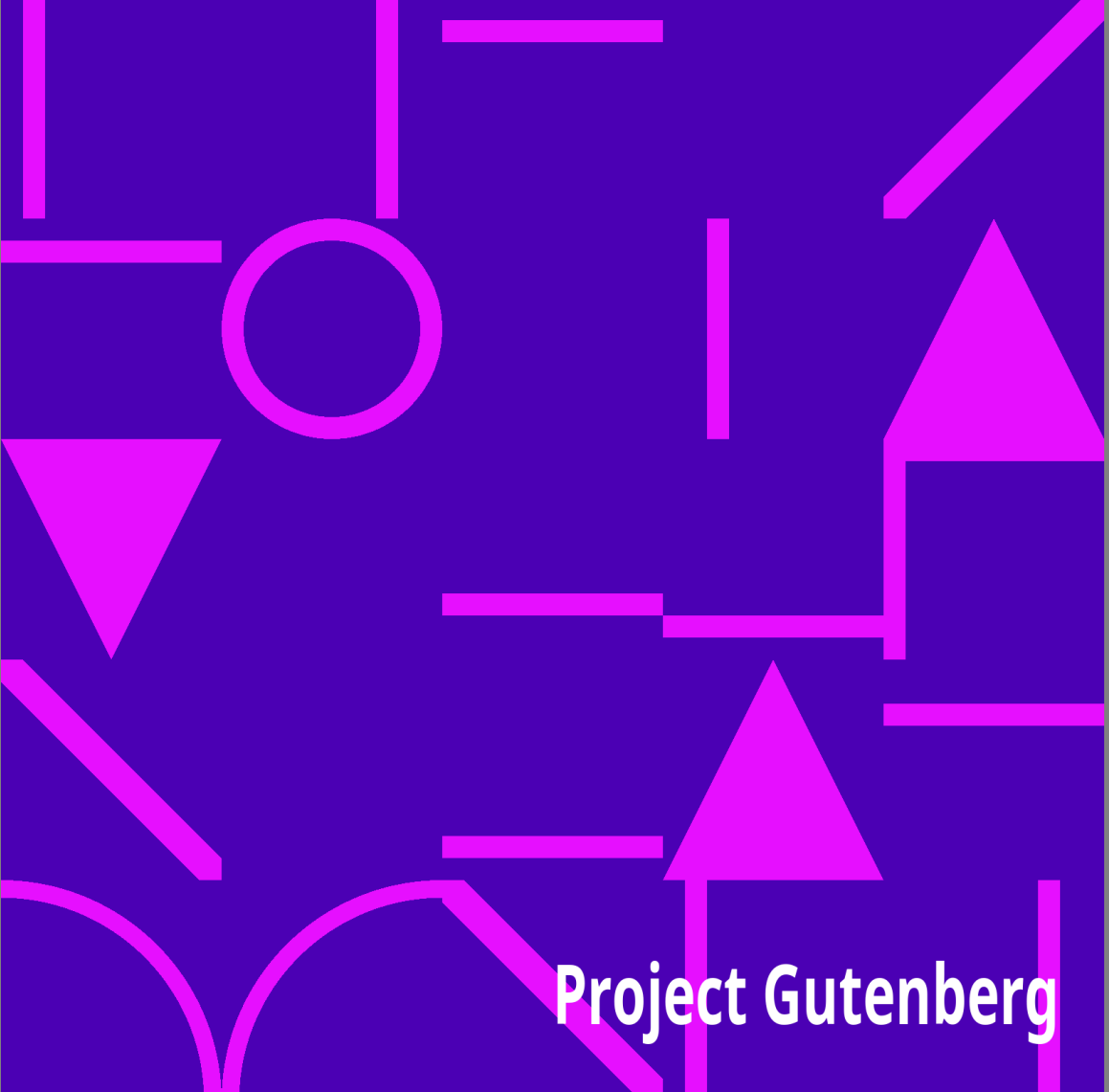
Ernest Rutherford and Harriet Brooks

An abstract geometric pattern consisting of various shapes and lines in shades of purple and magenta. The pattern includes a large circle, several triangles, and numerous straight lines of varying lengths and orientations, creating a complex, layered visual effect.

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The new gas from radium

Ernest Rutherford and Harriet Brooks

An abstract geometric pattern on a dark blue background, featuring various shapes in a lighter blue color. The shapes include circles, triangles, squares, and lines of different orientations and lengths, creating a complex, non-repeating design.

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*** START OF THE PROJECT GUTENBERG EBOOK THE NEW GAS
FROM RADIUM ***

IV.-The New Gas from Radium.

**By E. Rutherford, M.A., D.Sc.
Macdonald Professor of Physics, McGill
University, Montreal;**

AND

Miss H. T. Brooks. M.A.

(Read May 23, 1901.)

In a recent number of the Comptes Rendus, an account was given by Curie of the evidence of the existence of a new gas from radium, which possesses remarkable physical properties. A specimen of very radioactive radium was placed in a glass vessel connected with a mercury pump. On exhausting to a low vacuum and allowing the apparatus to stand, the pressure steadily increased. When the very small volume of gas thus collected flowed along the glass tubes, it rendered them phosphorescent, and if left in for some time, rapidly blackened them. The gas itself was powerfully radioactive, *i.e.* it continuously gave out a type of Röntgen rays, which made gases partial conductors of electricity and rapidly acted on a photographic plate. This gas preserves its radioactive power for several weeks.

For some time past, one of the authors had been independently investigating one of the most remarkable properties of radioactive substances, namely, the power of continuously emitting radioactive particles of some kind. The term "emanation" was applied to the substance thus emitted, as there was no evidence at the time whether the material emission was a vapour of the substance, a radioactive gas, or particles of matter each containing a large number of molecules.

The "emanation" from thorium compounds was shown to retain its radioactivity for several minutes and possessed the remarkable property of causing every substance in the neighbourhood of the thorium to become itself radioactive for several days. The specimens of impure radium then in the possession of the author, did not possess the power of emitting such an emanation; but Dorn, using a later and more active preparation of radium, showed that it possessed the same emanating power as thorium. One of the most interesting properties of excited radioactivity is that it can be concentrated in an electric field on the kathode, so that a very fine wire of any metal can be made to act like a powerfully radioactive substance for several days.

A short time ago, one of the authors published an account of the effect of temperature on the emanating power of radioactive substances, in the *Physikalische Zeitschrift*. In the paper it was shown that the emanating power of thorium increased with rise of temperature to about a red heat, but on heating to a white heat the emanating power was destroyed and could not be recovered. An examination of a specimen of radium obtained from De Haen, Hanover, showed that the effect of temperature on its emanating power was very large. When the substance was heated below a red heat, its emanating power increased over 10,000 times, but was to a large extent destroyed by heating to a higher temperature. The emanation, obtained by passing a slow current of air over heated radium, was found to preserve its radioactive powers for weeks, when kept in a closed metal vessel. The radioactivity slowly decayed, but was still quite appreciable after a month's interval.

The question now arose, if any physical experiments could be devised to settle the problem as to whether the emanation was in reality a radioactive gas, driven off from the substance, or a vapour of the substance, or a material emission of particles much larger than molecules. Experiments on thorium showed that no appreciable volume of gas could be collected by leaving thorium oxide in a vacuum tube connected with a mercury pump. No new lines were observed in the spectrum of the gas. The amount of the emanation thus given off was thus too small to detect by its volume in this way, but the electrical conductivity produced by the emanation in the gas, with which it is mixed, is often very large and can be used as a measure of the amount of emanation present. The emanation gives out a type of

radiation which ionizes the surrounding gas. When a strong electric field is applied, the current through the gas reaches a maximum value, and is then a measure of the total number of ions produced per second, multiplied by the charge on the ion.

By determining the rate of diffusion of the emanation into air or other gases, using the electrical method, it is possible to obtain an approximate estimate of its molecular weight. The coefficient of inter-diffusion of most of the simple gases have long been known, and the results show the coefficient of diffusion of one gas into another is inversely proportional to the square root of the product of the molecular weights. If therefore the coefficients of diffusion of the emanation into air is found to have a value lying between that of two gases A and B , we can conclude that the molecular weight of the emanation lies between the molecular weights of A and B .

The apparatus employed was similar to that used by Loschmidt^[1] in his experiments on the coefficient of inter-diffusion of gases in the year 1871.

Fig. (1) shows the general arrangement. A long brass cylinder AB 6 cms . in diameter, 73 cms. long, was divided into two equal parts by a movable metal slide S . The ends of the cylinder were closed with ebonite stoppers. Two insulated brass rods a and b , each half the length of the tube, passed through the ebonite stoppers and were supported centrally in the tube. The cylinder was insulated and connected to one pole of a battery of 300 volts, the other pole of which was to earth. The central rods could be connected to a sensitive quadrant electrometer.

The cylinder was covered with a thick layer of felt, and placed inside a metal box filled with cotton wool, in order to keep temperature conditions as steady as possible.

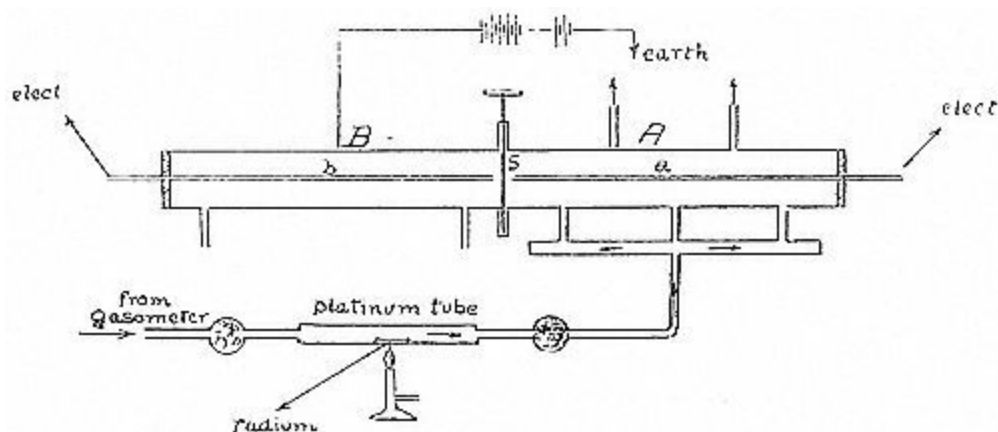


Fig. I.

In order to carry a sufficient quantity of emanation into the half cylinder *A*, it was necessary to slightly heat the radium. The slide *S* was closed and the side tubes opened. A slow current of dry air from a gas bag, passed through a platinum tube, in which a small quantity of a radium compound was placed. The emanation was carried with the air into the cylinder *A*. When a sufficient quantity had been introduced, as tested by the electrometer, the current of air was stopped. The side tubes were closed by fine capillary tubes. These prevented any appreciable loss of gas due to diffusion, but served to keep pressure of gas inside *A* at pressure of outside air. The three entrance tubes into the cylinder, shown in the figure, were for the purpose of initially mixing the emanation and gas as uniformly as possible.

After standing for several hours to make temperature conditions steady, the slide was opened, and the emanation began to diffuse into the tube *B*.

The current through the tubes *A* and *B* was measured by an electrometer, with suitable capacity in parallel, at regular intervals. Initially there is no current in *B*, but after the opening of the slide, the amount in *A* decreased and the amount in *B* steadily increased. After several hours the amount in each half is nearly the same, showing that the emanation is nearly uniformly diffused throughout the cylinder.

It can be readily shown that if

K = coefficient of diffusion of the emanation into air.

f = duration of diffusion experiments in secs.

a = total length of cylinder.

S = amount of emanation in tube A at end of diffusion.

S_2 = amount of emanation in tube B at end of diffusion, then

$$\frac{S_1 - S_2}{S_1 + S_2} = \frac{8}{\pi^2} \left\{ e^{\frac{-\pi^2 K t}{a^2}} + \frac{1}{9} e^{\frac{-\pi^2 K t}{a^2}} + \text{etc.} \right\}$$

SEE STEFAN AND LOSCHMIDT, BERICHTE WIEN. AKAD., 63, 1871.

From this equation K can be determined, if S_1 and S_2 are known.

An uncertainty however arises in estimating S_1 and S_2 for the rate of leak in A and B is made up of the current due to emanation alone and the current produced in the gas by the excited radioactivity on the electrodes. As the amount of excited radioactivity increases with the time, the ratio of the current due to the emanation and to the excited radiation varies with the time allowed for diffusion. The ratio of the current due to the excited radiation can be determined by removing the central electrode and finding the amount of current immediately after the introduction of a new electrode.

When the emanation is allowed to diffuse for half an hour, the current due to excited radioactivity was about $\cdot 4$ of the whole.

The calculated value of K was found to be about 20 per cent greater when the correction for the amount of excited radioactivity was applied.

The value of K deduced from the experiments was found to be between $\cdot 08$ and $\cdot 15$. All the later observations gave a value about $\cdot 08$.

This variation in the value of K deduced from the experiments is not altogether due to errors of experiment, the values obtained at first with a new specimen of radium were in all cases higher than when it had been laid by for several months. It appears as if the emanation were not simple in character, and that part of the emanation first given off was of lower molecular weight than that emitted after several months exposure to the air. Further experiments are now being carried out to see if the radium emanation undergoes a progressive change with time. For the purpose of comparison, we will now give a few of the coefficients of inter-diffusion of gases, compiled from Landölt and Bernstein's tables.

Gas or vapour	Coefficient of diffusion into air.	Molecular weight
Water vapour	0 · 198	18
Carbonic acid gas	0 · 142	44
Alcohol	0 · 101	46
Ether	0 · 077	74

In the above table we see that the coefficient of inter-diffusion follows the inverse order of the molecular weights. In cases of the simpler gases it has been shown experimentally that the coefficient of inter-diffusion is approximately inversely proportional to the square root of the product of the molecular weight. If we apply these considerations to the emanations ($K = \cdot 08$ to $\cdot 15$) we see that it is a gas or a vapour of molecular weight (allowing a wide margin) probably lying between 40 and 100. These numbers exclude the possibility of the substance being a vapour of radium, for it has already been shown by M. and Mme. Curie that the atomic weight of radium is greater than that of barium. We must therefore conclude that the emanation is in reality a heavy radioactive vapour or gas.

On account of the rapid decay of the radiating power of thorium emanations, it is not possible to determine its coefficients of diffusion in the same way; but special experiments show that it diffuses rapidly, and is also probably gaseous in character. The physical properties of these emanations or gases are most remarkable. The radium emanation not only continues for long intervals to be a source of radiation which is apparently similar in character to easily absorbed Röntgen rays, but in some way manufactures

from itself a positively charged substance, which travels to the negative electrode and becomes a source of secondary radioactivity.

FOOTNOTES:

[\[1\]](#) Wiener Akad., 1871.

TRANSCRIBER'S NOTES

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FROM RADIUM ***

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